Chapter 3 Multidimensional Random Variables

Section 5 Multidimensional Random Variables

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Outline

1. Multidimensional Random Variables
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   - n-Dimensional Discrete and Continuous Variables
   - Marginal Distributions
   - Independence
   - Functions of n-Dimensional Random Variables
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Definition: n-Dimensional Random Variable

Definition

Let $X_1, X_2, \ldots, X_n$ be n random variables defined on the same sample space Ω. The ordered tuple

$$(X_1, X_2, \ldots, X_n)$$

is called an **n-dimensional random variable** (or **n-dimensional random vector**).

Intuition: Just as a 2D random variable maps $\Omega \to \mathbb{R}^2$, an n-dimensional random variable maps $\Omega \to \mathbb{R}^n$.

Definition: Joint Distribution Function

Definition

Let $(X_1, \ldots, X_n)$ be an n-dimensional random variable. For any real numbers $x_1, x_2, \ldots, x_n$, the function

$$F(x_1, x_2, \ldots, x_n) = P(X_1 \leq x_1,\ X_2 \leq x_2,\ \ldots,\ X_n \leq x_n)$$

is called the **joint distribution function** of $(X_1, \ldots, X_n)$.

Properties of the n-Dimensional Joint CDF

Similar to the 2D case, the n-dimensional joint CDF $F(x_1, \ldots, x_n)$ satisfies:

- **Monotonicity:** Non-decreasing in each argument.
- **Boundedness:** $0 \leq F \leq 1$, with limits to 0 and 1 at the extremes.
- **Right-continuity:** Right-continuous in each argument.

n-Dimensional Rectangle Probability Formula

For $a_i < b_i$ $(i = 1, \ldots, n)$:

$$P(a_1 < X_1 \leq b_1, \ \ldots, \ a_n < X_n \leq b_n) = \Delta_{a_1}^{b_1} \cdots \Delta_{a_n}^{b_n} F \geq 0,$$

where $\Delta_{a_i}^{b_i} F = F(\ldots, b_i, \ldots) - F(\ldots, a_i, \ldots)$.

n-Dimensional Discrete Random Variable

Definition

If each component of $(X_1, \ldots, X_n)$ is a discrete random variable, then it is called an **n-dimensional discrete random variable**.

Joint PMF

Let the possible values be $(x_{1i_1}, x_{2i_2}, \ldots, x_{ni_n})$. The function

$$p_{i_1 i_2 \ldots i_n} = P(X_1 = x_{1i_1},\ X_2 = x_{2i_2},\ \ldots,\ X_n = x_{ni_n})$$

is the **joint PMF**.

Properties:

$$p_{i_1 i_2 \ldots i_n} \geq 0, \qquad \sum_{i_1} \sum_{i_2} \cdots \sum_{i_n} p_{i_1 i_2 \ldots i_n} = 1.$$

n-Dimensional Continuous Random Variable

Definition

If there exists a non-negative integrable function $f(x_1, \ldots, x_n)$ such that

$$F(x_1, \ldots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(u_1, \ldots, u_n) du_n \ldots du_1,$$

then it is an n-**dimensional continuous random variable**, and f is the **joint probability density function**.

Properties:

1. **Non-negativity:** $f(x_1, \ldots, x_n) \geq 0$.
2. **Normalization:** $\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} f(x_1, \ldots, x_n) dx_1 \cdots dx_n = 1$.
3. **Region Probability:** For any region $G \subset \mathbb{R}^n$, $P((X_1, \ldots, X_n) \in G) = \iint \cdots \int_G f(x_1, \ldots, x_n) dx_1 \cdots dx_n$.

Marginal Distribution Function

Definition: Marginal Distribution

If we only care about a subset of variables, say X_1, we let the other $n-1$ variables go to $+\infty$:

$$F_{X_1}(x_1) = F(x_1, +\infty, \ldots, +\infty) = \lim_{x_2 \to +\infty} \cdots \lim_{x_n \to +\infty} F(x_1, x_2, \ldots, x_n).$$

Similarly, for any k components $(X_1, \ldots, X_k)$:

$$F_{X_1,\ldots,X_k}(x_1, \ldots, x_k) = F(x_1, \ldots, x_k, +\infty, \ldots, +\infty).$$

Physical Meaning: We no longer care about the specific values of the other variables, so we let them take all possible values (approach $+\infty$) to "marginalize" them out.

Discrete and Continuous Marginals

Discrete Case:
If $(X_1, \ldots, X_n)$ is discrete with joint PMF $p_{i_1 \ldots i_n}$, the marginal PMF of X_1 is:

$$p_{1i_1} = P(X_1 = x_{1i_1}) = \sum_{i_2} \cdots \sum_{i_n} p_{i_1 i_2 \ldots i_n}.$$

"Fix one index, sum over all others."

Continuous Case:
If $(X_1, \ldots, X_n)$ is continuous with joint density $f(x_1, \ldots, x_n)$, the marginal density of X_1 is:

$$f_{X_1}(x_1) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} f(x_1, x_2, \ldots, x_n) dx_2 \cdots dx_n.$$

For the first k components, integrate out the remaining $n - k$ variables.

Independence of n Random Variables

Definition

Let $X_1, X_2, \ldots, X_n$ be n random variables with joint CDF $F(x_1, \ldots, x_n)$ and marginal CDFs $F_{X_i}(x_i)$. If for any real numbers $x_1, \ldots, x_n$,

$$F(x_1, x_2, \ldots, x_n) = F_{X_1}(x_1) F_{X_2}(x_2) \cdots F_{X_n}(x_n),$$

then $X_1, X_2, \ldots, X_n$ are **mutually independent**.

Equivalent Conditions:

- **Discrete:** $p_{i_1 i_2 \ldots i_n} = p_{1 i_1} p_{2 i_2} \cdots p_{n i_n}$ for all indices.
- **Continuous:** $f(x_1, \ldots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$ for all $x_1, \ldots, x_n$.
- **Quick Test:** The support of the joint density is an n-dimensional "box" (not slanted or curved), and the density factors into single-variable functions.

Remark: Independent and Identically Distributed (i.i.d.) Sequence

i.i.d. Sequence

Let $X_1, X_2, \ldots$ be an infinite sequence of random variables.

- If for any positive integer n, $X_1, \ldots, X_n$ are mutually independent, the sequence is **independent**.
- If in addition they all have the same distribution $F_{X_1}(x) = F_{X_2}(x) = \cdots = F(x)$, the sequence is **independent and identically distributed** (i.i.d.).

For a continuous i.i.d. sequence with common density $f(x)$, the joint density of n variables is:

$$f(x_1, \ldots, x_n) = f(x_1)f(x_2)\cdots f(x_n) = \prod_{i=1}^{n} f(x_i).$$

Application: This is the foundation of simple random samples in statistics: "A sample $X_1, \ldots, X_n$ from population X" means they are i.i.d. and share the same distribution as X.

Functions of n-Dimensional Random Variables

Let $Z = g(X_1, X_2, \ldots, X_n)$. The basic idea is the same as in the 2D case, but the calculation is more complex.

Discrete Case:

$$P(Z = z_k) = \sum_{g(x_{1i_1}, \ldots, x_{ni_n}) = z_k} p_{i_1 i_2 \ldots i_n}.$$

Continuous Case (CDF Method):

1. Write the CDF: $F_Z(z) = P(g(X_1, \ldots, X_n) \leq z)$.
2. Find the region $D_z = \{(x_1, \ldots, x_n) : g(x_1, \ldots, x_n) \leq z\}$.
3. Compute the multiple integral over D_z.
4. Differentiate with respect to z to get $f_Z(z)$.

Remark: Common Statistics

In statistics, the most commonly used functions of n-dimensional random variables are **statistics** constructed from samples:

- **Sample Mean:** $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$.
- **Sample Variance:** $S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$.
- **Maximum and Minimum:** $X_{(n)} = \max(X_1, \ldots, X_n)$, $X_{(1)} = \min(X_1, \ldots, X_n)$.

Proposition: Sum of Independent Normals

Proposition

If $X_1, X_2, \ldots, X_n$ are mutually independent and $X_i \sim N(\mu_i, \sigma_i^2)$, then their sum $Z = X_1 + \cdots + X_n$ follows a normal distribution:

$$Z \sim N\left(\sum_{i=1}^{n} \mu_i,\ \sum_{i=1}^{n} \sigma_i^2\right).$$

Note: This is a direct extension of the 2D case. The proof uses induction and the 2D convolution formula.

Proposition: Distribution of Maximum and Minimum

Proposition

If $X_1, \ldots, X_n$ are independent and have the same CDF $F(x)$, then the maximum $M = \max(X_1, \ldots, X_n)$ and minimum $N = \min(X_1, \ldots, X_n)$ have CDFs:

$$F_M(z) = [F(z)]^n, \qquad F_N(z) = 1 - [1 - F(z)]^n.$$

Special Case: If $X_i \sim \text{Exp}(\lambda)$, then $N \sim \text{Exp}(n\lambda)$.

- **Series System:** The lifetime is the minimum. With n components, the failure rate becomes $n\lambda$.
- **Parallel System:** The lifetime is the maximum.

Proposition: Chi-Square Distribution

Proposition

Let $X_1, X_2, \ldots, X_n$ be mutually independent and all follow the standard normal distribution $N(0,1)$. The sum of their squares

$$\chi^2 = X_1^2 + X_2^2 + \cdots + X_n^2$$

follows the **chi-square distribution** with n degrees of freedom, denoted as $\chi^2 \sim \chi^2(n)$. Its density function is:

$$f_{\chi^2}(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}, \quad x > 0.$$

Remark: Why is χ^2 Important?

- The χ^2 distribution is a direct extension of the normal distribution.
- It is the theoretical foundation for many statistical distributions, including the **t-distribution** and **F-distribution**.
- In hypothesis testing, the χ^2 test is used to determine goodness of fit and independence of categorical variables.

Note: The derivation requires advanced calculus (multiple integrals of spherical symmetry). Students only need to understand the result and its applications.

Section Summary

- **Joint Distribution:** Described by an n-dimensional CDF or joint density/PMF. It has monotonicity, boundedness, right-continuity, and a multi-dimensional rectangle formula.
- **Marginal Distribution:** Obtained by summing/integrating out the unwanted variables. Joint distribution determines marginals, but not vice versa.
- **Independence:** $F(x_1, \ldots, x_n) = \prod_{i=1}^{n} F_{X_i}(x_i)$.
- **Functions of RVs:** Use the CDF method. Key results: Sums of normals are normal; maximum/minimum distributions; sum of squared standard normals is χ^2.